MINIPROJECT REPORT_MPJ
(Title Page)

Project Title: Hotel Reservation System

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Submission Date: 08 /04 / 2026

Certificate (Optional)

This is to certify that the project titled “Hotel Reservation System” is successfully completed by under Prof. Suhas Joshi Sir guidance…

Acknowledgement

- Gratitude to guide, institution, sponsors, and team members

Abstract / Executive Summary

The Hotel Reservation System is a full-stack mobile application designed to streamline hotel booking operations through a modern Android interface and a robust Spring Boot backend. The system allows users to search for rooms based on availability, manage bookings, and perform real-time actions such as cancellation and Do Not Disturb (DND) requests. The application solves the problem of manual booking inefficiencies by providing a digital platform with real-time availability tracking and seamless user experience. The backend efficiently handles business logic, including booking validation, availability calculation, and refund processing. The system uses Kotlin for Android development, Spring Boot for backend services, and MySQL for data management. The backend is deployed on Render and the database on Railway, ensuring scalability and accessibility. The final outcome is a fully functional, user-friendly, and scalable hotel booking system with modern UI/UX design and real-time data synchronization.

1. Project Overview

Introduction

Hotel booking systems are essential in modern hospitality management. This project provides a digital solution for booking, managing, and tracking hotel reservations.

Purpose

To develop a real-time booking system with:

- Seamless user experience
- Accurate availability tracking
- Efficient booking management

2. Problem Statement

2.1 Business Problem

- Manual booking systems are inefficient
- No real-time availability
- Poor user experience

2.2 Technical Problem

- Lack of centralized database
- No dynamic room allocation
- No automation for booking management

3. Scope

Included:

- User authentication
- Room booking system
- Availability checking
- Booking management
- Room Upgradation Logic
- Room cancellation and booking discount for particular coupon

Excluded:

- Payment gateway
- Admin dashboard

4. Objectives

Primary:

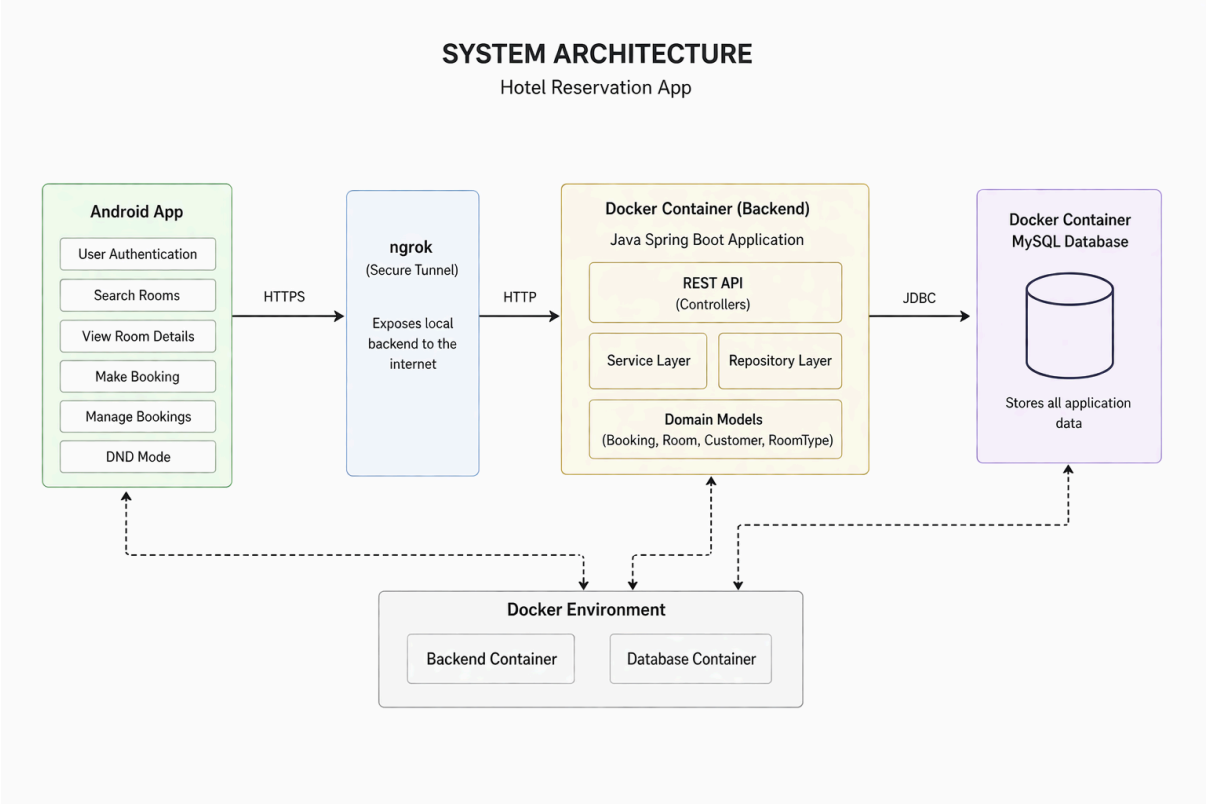
- Develop a full-stack hotel booking system

Secondary:

- Implement real-time availability
- Provide modern UI/UX
- Enable booking management

5. System Architecture & Tools Used

5.1 System Architecture



The Hotel Reservation System follows a Three-Tier Architecture, which separates the application into three logical layers: Presentation Tier, Middle Tier (API Layer), and Data Tier. This architecture improves scalability, maintainability, and performance by clearly dividing responsibilities.

1. Presentation Tier (Android Application):

The presentation layer is developed using Android (Kotlin) and is responsible for user interaction. It provides the graphical user interface through activities and fragments. Users can perform operations such as login, searching rooms, booking, and managing reservations. The app communicates with the backend using Retrofit, which handles HTTP requests and responses in JSON format. GSON is used for parsing JSON data, while RecyclerView and Material Design components are used to create a modern and responsive UI.

2. Middle Tier (API Layer – Spring Boot Backend):

The middle tier acts as the core processing unit of the system. It is implemented using Spring Boot and exposes RESTful APIs that handle all client requests.

This layer is divided into multiple sub-layers:

- Controller Layer: Handles incoming HTTP requests and maps them to appropriate services.

- Service Layer: Contains business logic such as room availability calculation, booking validation, and cancellation rules.

- Repository Layer: Manages data access using Spring Data JPA and Hibernate.

- Entity Layer: Represents database tables as Java objects.

- Exception Handling: Ensures proper error handling and response management.

The backend processes requests received from the Android application, interacts with the database, and returns appropriate responses.

3. Data Tier (MySQL Database):

The data tier consists of a MySQL database that stores all application data. It includes tables such as rooms, customer, bookings, and service_requests. This layer is responsible for persistent data storage and retrieval. The backend communicates with the database using JPA (Java Persistence API) and JDBC, ensuring efficient and secure data handling.

Working Flow:

The Android application sends HTTP requests to the Spring Boot backend in JSON format. The backend processes the request, applies business logic, interacts with the MySQL database, and sends the response back to the client. This ensures real-time data synchronization and accurate booking management.

Advantages of the Architecture:

- Separation of concerns
- Scalability and flexibility
- Easy maintenance and debugging
- Improved performance and security

5.2 Tools & Technologies

Category	Tools
Programming	Kotlin, Java
Backend	Spring Boot
Database	MySQL
API	REST API
Platform	Android Studio, Render

6. Project Work Distribution

Rank	Team Member	Role	Functions Implemented
1	Saaransh Johri	Backend Core Logic & Database	Implemented complex booking logic, availability calculation, database design, and business rules like cancellation and DND
2	Milind Muravane	API Integration	Handled communication between frontend and backend, managed Retrofit calls, JSON parsing, and debugging API errors
3	Aditya Kiran	Controllers & Exception Handling	Developed REST controllers, managed request handling, and implemented exception handling for stable backend responses
4	Tanmay Dhumal	Android UI/UX Development	Focused on UI design, layout implementation, and user experience improvements using modern design concepts

7. Explanation of Project Modules

The Hotel Reservation System is structured into multiple functional modules, each responsible for a specific aspect of the application. These modules collectively ensure efficient operation, seamless user interaction, and proper management of booking processes.

Module 1: User Authentication

Description:
This module manages user registration and login functionality, ensuring secure access to the system.

Functionality:

- User registration with details such as Name, Email, Phone, and Password
- User login authentication through backend APIs
- Session management using SharedPreferences

Inputs:

- User registration details
- Login credentials (Email, Password)

Outputs:

- Successful authentication or error response
- Active user session

Module 2: Room Search and Filtering

Description:
This module enables users to search for available rooms based on selected criteria.

Functionality:

- Selection of check-in and check-out dates
- Selection of number of adults and children
- Dynamic filtering of rooms based on availability

Inputs:

- Date range
- Guest count

Outputs:

- List of available rooms matching criteria

Module 3: Room Listing and Availability

Description:
This module displays available rooms along with relevant details.

Functionality:

- Displays room categories (Suite, Deluxe, Standard)
- Shows price per night
- Displays real-time availability of rooms

Inputs:

- Search parameters from user

Outputs:

- Detailed list of available rooms

Module 4: Booking Management

Description:
This module handles the process of booking rooms.

Functionality:

- Collection of guest details dynamically
- Application of coupon code (MATS – 5% discount)
- Booking confirmation and data storage

Inputs:

- Guest details
- Selected room

Outputs:

- Booking confirmation
- Stored booking record

Module 5: My Bookings (User Dashboard)

Description:
This module allows users to view and manage their bookings.

Functionality:

- Displays all bookings (past and upcoming)
- Shows booking status (BOOKED or CANCELLED)
- Displays booking details

Inputs:

- User ID

Outputs:

- List of bookings

Module 6: Do Not Disturb (DND) Control

Description:
This module allows users to manage room privacy during their stay.

Functionality:

- Toggle DND status for a booking
- Updates status in backend

Inputs:

- Booking ID

Outputs:

- Updated DND status

Module 7: Booking Cancellation

Description:
This module handles cancellation of bookings.

Functionality:

- Allows users to cancel bookings
- Applies 17% deduction logic
- Updates booking status

Inputs:

- Booking ID

Outputs:

- Updated booking status (CANCELLED)

8. Business Use Cases

This project addresses real-world challenges in hotel booking and management by providing a digital and automated solution.

Use Case 1: Online Room Booking

- Problem: Customers face difficulty in booking rooms due to manual processes and lack of real-time availability.
- Solution: The system allows users to search and book rooms based on date and occupancy in real-time.
- Benefit: Reduces manual effort and ensures accurate booking.

Use Case 2: Booking Management

- Problem: Managing bookings manually leads to errors and inefficiency.
- Solution: The system provides a centralized platform to view, manage, and cancel bookings.
- Benefit: Improves operational efficiency and reduces errors.

Use Case 3: Room Availability Tracking

- Problem: Overbooking or incorrect availability information.
- Solution: Backend logic dynamically calculates room availability.
- Benefit: Ensures reliable and real-time availability data.

9. Application Use Cases

Use Case 1: User Login

- Actors: User
- Workflow:
    1. User enters email and password
    2. System validates credentials
    3. User is authenticated
- Expected Outcome:
  User successfully logs into the system

Use Case 2: Room Search

- Actors: User
- Workflow:

1. User selects check-in and check-out dates
2. User selects number of guests
3. System fetches available rooms

- Expected Outcome:
  List of available rooms is displayed

Use Case 3: Booking a Room

- Actors: User
- Workflow:
  1. User selects a room
  2. Enters guest details
  3. Applies coupon (optional)
  4. Confirms booking
- Expected Outcome:
  Booking is successfully created

Use Case 4: Cancel Booking

- Actors: User
- Workflow:
  1. User navigates to “My Bookings”
  2. Selects a booking
  3. Clicks cancel
- Expected Outcome:
  Booking status updated to CANCELLED

10. Future Scope

- Integration of online payment gateway
- Development of admin dashboard for hotel staff
- Implementation of push notifications
- Integration with third-party travel platforms

- Scalability for multi-hotel support

11. Challenges Faced

Technical Challenges

- Handling real-time room availability logic
- Managing API integration between Android and backend
- Resolving database crashes and inconsistencies

Team/Management Challenges

- Coordination between frontend and backend development
- Synchronizing API structure with UI requirements

Resource Limitations

- Limited time for advanced feature implementation
- Lack of production deployment environment

Resolution

- Modular development approach adopted
- Frequent testing and debugging performed
- Proper API structuring ensured smooth integration

12. Outcome

- Successfully developed a complete hotel reservation system
- Real-time booking and availability implemented
- Smooth communication between frontend and backend

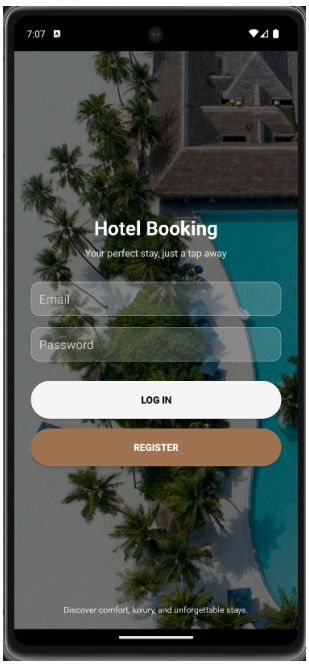
Performance Improvements

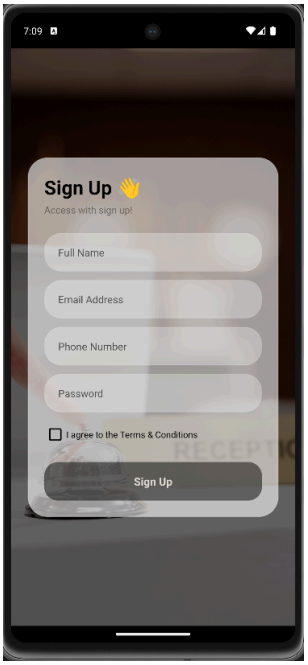
- Faster booking process compared to manual system
- Reduced chances of booking errors

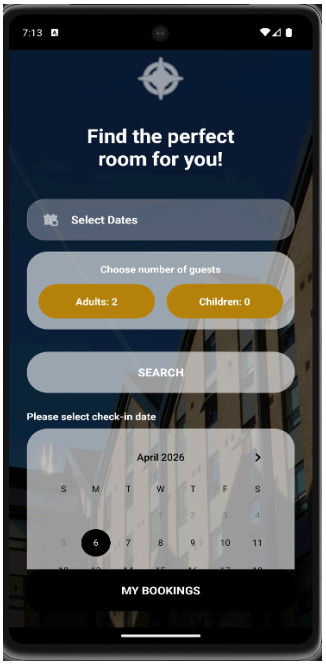
Metrics (if applicable)

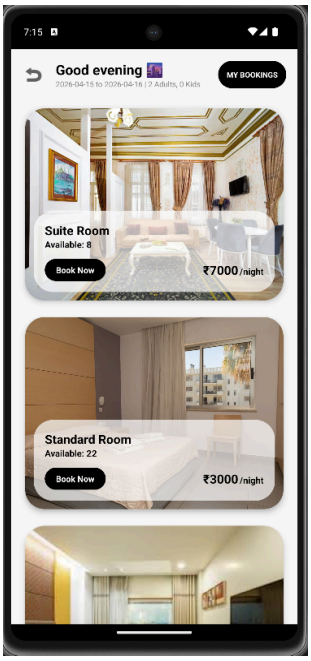
- API response time within acceptable limits
- Successful execution of all major functionalities

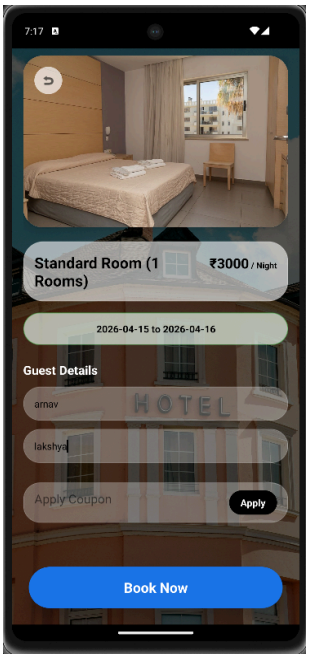
Screenshots (Recommended)

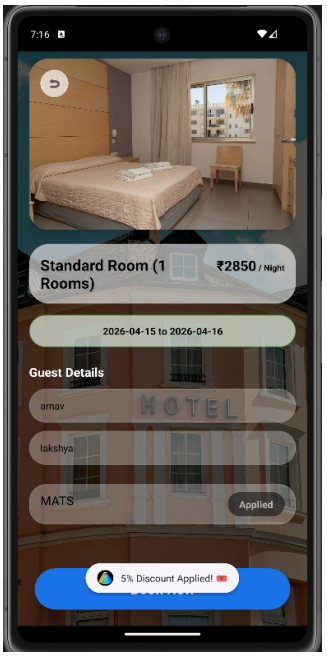

7:07
Hotel Booking
Your perfect stay, just a tap away
Email
Password
LOG IN
REGISTER
Discover comfort, luxury, and unforgettable stays.

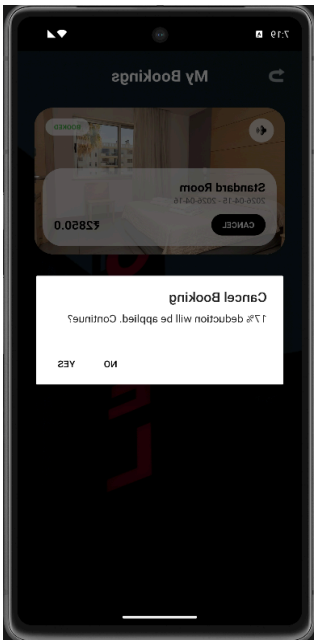

7:09
Sign Up
Access with sign up!
Full Name
Email Address
Phone Number
Password
I agree to the Terms & Conditions
Sign Up

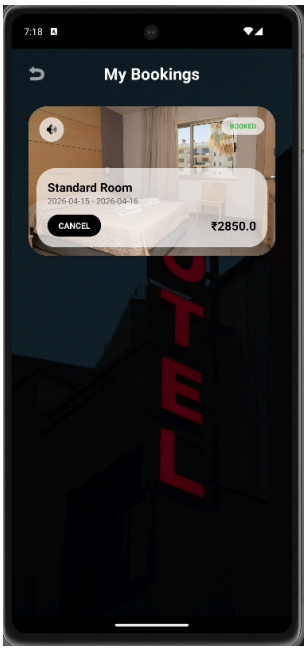

7:13
Find the perfect room for you!
Select Dates
Choose number of guests
Adults: 2
Children: 0
SEARCH
Please select check-in date
April 2026
S M T W T F S
MY BOOKINGS

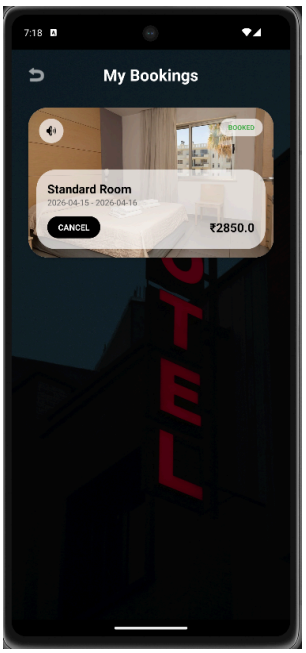

7:15
Good evening
2026-04-15 to 2026-04-16 | 2 Adults, 0 Kids
MY BOOKINGS
Suite Room
Available: 8
Book Now
₹7000 /night
Standard Room
Available: 22
Book Now
₹3000 /night

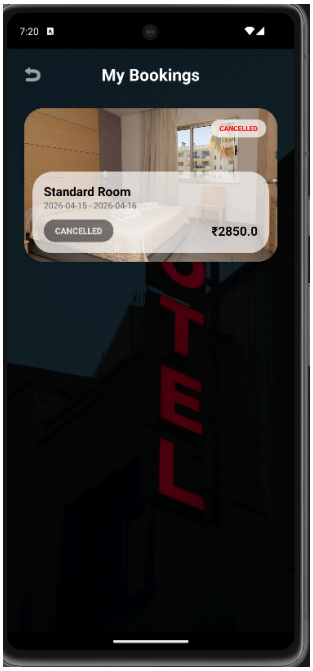

7:17
Standard Room (1 Rooms)
₹3000 / Night
2026-04-15 to 2026-04-16
Guest Details
arnav
lakshya
Apply Coupon
Apply
Book Now


7:16
Standard Room (1 Rooms)
₹2850 / Night
2026-04-15 to 2026-04-16
Guest Details
arnav
lakshya
MATS
Applied
5% Discount Applied!









13. Conclusion

The Hotel Reservation System effectively demonstrates the integration of mobile application development with backend services. The system provides a user-friendly interface and efficient booking management.

Learning Outcomes

- Understanding of Android development using Kotlin

- Experience with REST API integration
- Knowledge of Spring Boot backend architecture
- Database design and management using MySQL

Final Impact

The system improves booking efficiency, reduces manual errors, and enhances user experience.

14. References

- **Android Developer Documentation –** https://developer.android.com
- **Spring Boot Documentation –** https://spring.io/projects/spring-boot
- **MySQL Documentation –** https://dev.mysql.com/doc/
- **Retrofit Documentation –** https://square.github.io/retrofit/
- **Material Design Guidelines –** https://material.io
- **UI References –** https://dribbble.com

Tools used:

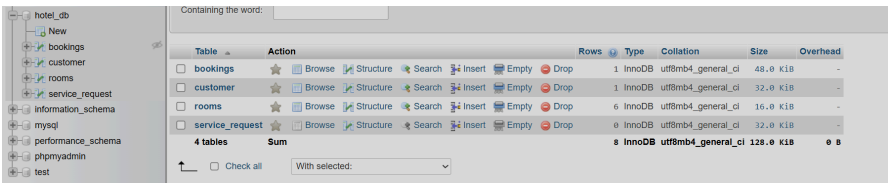
- **Android Studio Agent**
- **Logcat for debugging**
- **LLMs**

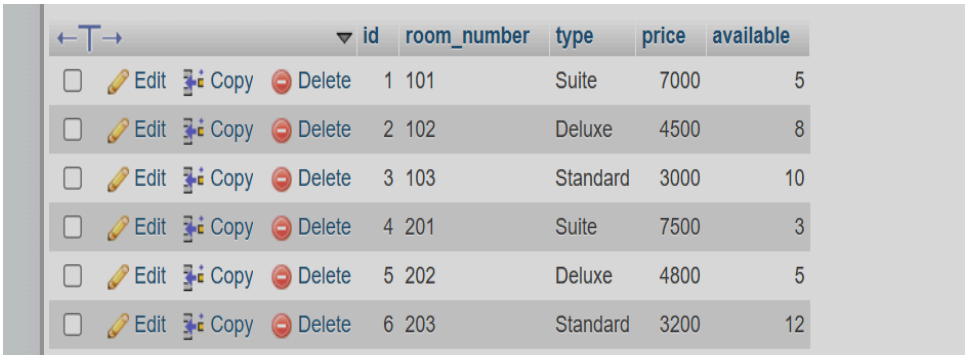
15. Appendix (Optional)

Database Tables

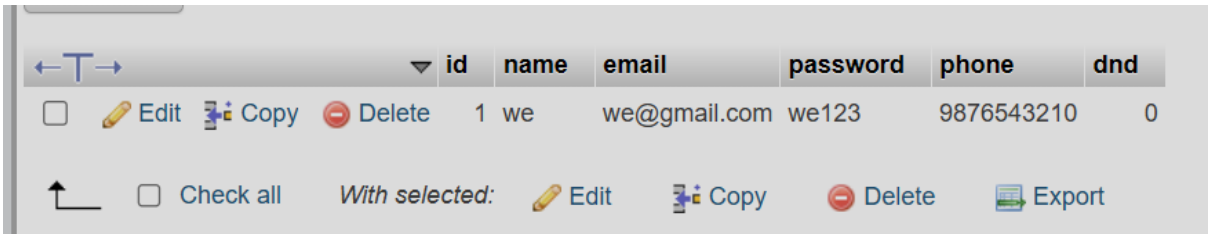
The following tables were used in the Hotel Reservation System:

- **Customer Table:** Stores user details such as name, email, phone number, and password.
- **Rooms Table:** Contains information about room types, pricing, and availability.
- **Bookings Table:** Stores booking details including customer, room, dates, status, and DND option.
- **Service_Request Table:** Maintains additional service requests made by users during their stay.

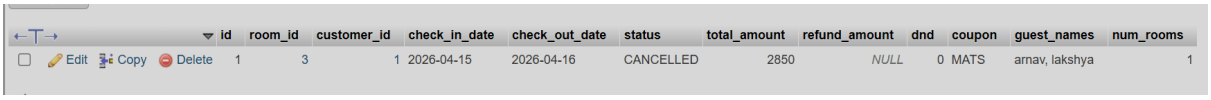




id	room_number	type	price	available
1	101	Suite	7000	5
2	102	Deluxe	4500	8
3	103	Standard	3000	10
4	201	Suite	7500	3
5	202	Deluxe	4800	5
6	203	Standard	3200	12



id	name	email	password	phone	dnd
1	we	we@gmail.com	we123	9876543210	0



id	room_id	customer_id	check_in_date	check_out_date	status	total_amount	refund_amount	dnd	coupon	guest_names	num_rooms
1	3	1	2026-04-15	2026-04-16	CANCELLED	2850	NULL	0	MATS	arnav, lakshya	1

Presentation (PPT) Structure

Use this to align your slides with the report:

1. Title Slide
2. Problem Statement
3. Objectives
4. System Architecture
5. Tools Used
6. Modules Overview
7. Use Cases

8. Demo / Screenshots
9. Challenges
10. Future Scope
11. Conclusion